

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396403
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Cooley; Devin

Ejection chute for agricultural baler

Abstract

An agricultural baler includes a bale chamber having a discharge outlet and a bale ejection system configured to discharge bales from the bale chamber on the ground surface of the field. The bale ejection system includes a chute with a first half portion and a second half portion that receives a first bale and a second bale discharged simultaneously from the discharge outlet in the first half portion and the second half portion, respectively, and an ejection actuator that pushes the second bale from the second half portion of the chute to the first half portion of the chute.

Inventors:	Cooley; Devin (Reading, PA)
Applicant:	CNH Industrial America LLC (New Holland, PA)
Family ID:	1000008779332
Assignee:	CNH Industrial America LLC (New Holland, PA)
Appl. No.:	18/211413
Filed:	June 19, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240415064 A1	Dec. 19, 2024

Publication Classification

Int. Cl.:	A01F15/08 (20060101)
U.S. Cl.:	
CPC	A01F15/0875 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5742010	12/1997	Griffin	177/253	G01G 19/08
7127985	12/2005	Standlee	53/399	A01F 15/0875
7895944	12/2010	Derstine	100/246	A01F 15/005
8915182	12/2013	Van Groenigen	N/A	N/A
8978550	12/2014	Demon	199/7	A01F 15/0875
10058037	12/2017	Kraus	N/A	A01F 15/145
2014/0216279	12/2013	Van Groenigen	100/4	A01F 15/0875
2018/0020621	12/2017	Jones	56/341	A01F 15/0705
2021/0227754	12/2020	Pramod	N/A	A01F 15/0705
2023/0180668	12/2022	Russell	56/341	A01F 15/14

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0650664	12/1994	EP	N/A

Primary Examiner: Crawford; Gene O

Assistant Examiner: Awais; Muhammad

Attorney, Agent or Firm: Buchanan Ingersoll & Rooney PC

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to agricultural balers, and, more particularly, to ejection systems used with square agricultural balers.

BACKGROUND OF THE INVENTION

(2) Agricultural harvesting machines, such as balers, for example, are used to consolidate and package crop material so as to facilitate the storage and handling of the crop material for later use. In the case of hay, a mower-conditioner cuts and conditions the crop material for windrow drying in the sun. In the case of straw, an agricultural combine discharges non-grain crop material from the rear of the combine defining the straw (such as wheat or oat straw, for example) which is to be picked up by the baler. The cut crop material is typically raked and dried, and a baler, such as a round baler or a square baler, for example, straddles the windrows and travels along the windrows to pick up the crop material and form it into round or square bales. More specifically, a pickup unit at the front of the baler gathers the cut and windrowed crop material from the ground and then conveys the cut crop material into a bale-forming chamber within the baler where the crop material is compacted, typically by means of a reciprocating plunger. The bale-forming chamber usually includes a device for tying bales and a discharge outlet, for example connected to a discharge chute for gently lowering bales onto the field. During normal baling operation, tied bales are ejected from

the baler through action of the plunger.

(3) Square agricultural balers are sometimes preferred because the square-shaped bales facilitate stacking, delivery, and use. During baling, a small square baler produces two small square bales simultaneously, and there is a need to align the ejected bales consecutively in a single row. If the bales are not aligned in a single row, an operator with a bale bundler cannot drive in a straight line to pick up the small square bales, but instead, will have to swerve back and forth to pick up bales that are ejected side by side. It is, therefore, inconvenient if the two simultaneously ejected bales land side by side after being ejected from the bale-forming chamber.

(4) What is needed in the art is an agricultural baler with an ejection system that can position simultaneously ejected bales consecutively along the same path in the field.

SUMMARY OF THE INVENTION

(5) Described herein is an improved ejection system for agricultural balers. The ejection system ensures that simultaneously ejected bales are positioned in the field consecutively along the same path by ejecting one bale at a time.

(6) An agricultural baler includes an ejection system that ejects bales that are discharged simultaneously from the bale-forming chamber one bale at a time. Although both bales slide from the bale-forming chamber onto the chute at the same time, only the first bale continues to slide directly off the chute and is ejected onto the ground surface of the field due to the chute being angled downward. The second bale slides down off the chute, but is stopped by a wall, ridge or lip formed at the bottom end of the chute. After the first bale has been ejected from the chute onto the ground surface of the field, an ejection actuator pushes the second bale in the lateral direction and slides it over to the side of the chute that does not have a wall, ridge or lip formed at the bottom end of the chute. The ejection actuator then retracts, thereby allowing the second bale to freely slide off the chute, like the first bale. This operating sequence results in the placement of the two ejected bales consecutively along the same path on the field.

(7) A method for ejecting bales that are discharged simultaneously from the bale-forming chamber one bale at a time is also provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawing figures depict one or more implementations, by way of example only, not by way of limitations. In the figures, like reference numerals refer to the same or similar elements.

(2) FIG. 1 illustrates an embodiment of a square agricultural baler to which a bale ejection system according to embodiments described herein may be applied;

(3) FIG. 2 is a top plan view of a bale ejection system of the baler of FIG. 1, which is shown schematically, according to an embodiment;

(4) FIG. 3 is a top plan view of the bale ejection system of FIG. 2 with an actuated ejection actuator.

(5) FIG. 4 is a side elevation view of a bale ejection system.

(6) FIG. 5 is a schematic diagram of an exemplary hydraulic control circuit for use with the bale ejection system according to an embodiment.

(7) FIG. 6 is a schematic diagram of an exemplary control circuit for use with the bale ejection system according to an embodiment.

(8) FIG. 7 is a schematic view of the positioning of bales in a field ejected with the bale ejection system according to an embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(9) The terms “forward,” “rearward,” “upward,” “downward,” “left,” and “right,” when used in connection with the agricultural baler described herein and/or components thereof are usually

determined with reference to the direction of forward operative travel of the towing vehicle and the height of the baler, but they should not be construed as limiting. The terms “longitudinal” and “transverse” are determined with reference to the fore-and-aft direction of the towing vehicle and the width of the baler, and are equally not to be construed as limiting.

(10) Referring now to the drawings, and more particularly to FIG. 1, illustrated is an agricultural baler **10**, which can be connected to and pulled behind an agricultural vehicle, such as a tractor (not shown), for example. The baler **10** includes a frame **25** which is equipped with a forwardly extending tongue **26** at its front end with hitch means (not shown) for coupling the baler **10** to a towing tractor (shown in FIG. 7).

(11) The baler **10** may operate on a two-stage feeding system. A pick-up assembly **27** lifts windrowed crop material off the field as the baler **10** is being pulled by a tractor, and delivers such material into the front end of a rearwardly and upwardly curved, charge-forming feeder duct **28**. The duct **28** communicates at its upper end with an overhead, fore-and-aft extending bale-forming chamber **11** into which crop charges are loaded by a cyclically operating stuffer mechanism **29**. A continuously operating packer mechanism **30** at the lower front end of the feeder duct **28** continuously feeds and packs material into the duct **28** as to cause charges of the crop material to take on and assume the internal configuration of the duct **28** prior to periodic engagement by the stuffer **29** and insertion up into the bale-forming chamber **11**. Each action of the stuffer mechanism **29** introduces a “charge” or “flake” of crop material from the duct **28** into the chamber **11**. The packer mechanism **30** functions as the first stage for crop material compression.

(12) The crop material can be transported into the bale-forming chamber **11** of the baler **10** with a conveyor, such as a rotating conveying rotor, for example. A plunger **13** reciprocates in a fore-and-aft direction ‘D’ within the bale-forming chamber **11**. Biomass fed via the feeder duct **28** is thereby compacted, e.g., compressed or otherwise treated, so as to form rectangular bales in the above-described operation of the agricultural baler **10**. These carrier elements are movable so that the bale-forming chamber **11** can initially contract and subsequently expand to maintain an appropriate amount of pressure on the periphery of the bale. The plunger **13** compresses the wads of crop material into flakes to form a rectangular or square bale and, at the same time, gradually advances the bale toward a discharge outlet **14** of the bale-forming chamber **11**, from where the bales may eventually be ejected. Together, the plunger **13** and the bale-forming chamber **11** function as the second stage for crop compression.

(13) The baler **10** may include components, such as a tying assembly or a knotter mechanism for automatically wrapping and tying the completed bales with a tie twine or similar lineal object, to make the bales self-supporting, for example for shipping and storage. When enough flakes have been added and the bale reaches a full (or other predetermined) size, the tying assembly is actuated in order to wrap and tie twine around the bales while the bales are still in the bale-forming chamber **11**. Once tied, the bales are discharged from the discharge outlet **14** of the bale-forming chamber **11** onto a discharge structure in the form of a chute **31**, for gently lowering the bales near ground level onto the field.

(14) Further details of baler **10** may be described in U.S. Patent App. Pub. No. 2013/0019765, which is incorporated by reference herein in its entirety and for all purposes.

(15) In a first aspect, an ejection system **200** is provided for an agricultural baler **10**, more particularly a small square baler or a rectangular baler, i.e., a machine for forming square or rectangular bales from agricultural crop material. In a small square baler, for example, the bales are discharged simultaneously from the discharge outlet **14** of the bale-forming chamber **11**, usually with the two bales positioned next to each other or in parallel relative to the side walls of the baler **10**.

(16) The ejection system **200** according to embodiments of the present invention may be a separate structure for being added or retro-fitted into an existing agricultural baler. Alternatively, the ejection system **200** may be built into an agricultural baler.

(17) According to an aspect of the present invention, and referring now specifically to FIG. 2, a bale ejection system **200** includes bale-forming chamber **11** where formed square bales **33** are arranged in multiple rows of two bales **33** positioned next to each other in each row. A chute **31** (see also FIG. 1) is operatively connected to the discharge outlet **14** of the bale-forming chamber **11** and simultaneously receives bales **33** as the bales **33** are discharged from the discharge outlet **14** of the bale-forming chamber **11**. The chute **31** includes two half portions **31a** and **31b** (illustrated in FIG. 2 as being separated by an imaginary dashed line) each of which extend downwardly at an angle from the discharge outlet **14** of the bale-forming chamber **11** toward the ground surface of the field. Alternatively, only half portion **31a** may extend downwardly at an angle, and half portion **31b** may be oriented horizontally (i.e., not angled downward). Two bales **33a** and **33b** slide from the bale-forming chamber **11** onto the chute **31** at the same time and substantially parallel to each other. Each of the two bales **33a** and **33b** slides out from the bale-forming chamber **11** in the forward and downward direction D1 and D2, for example, onto a corresponding half portion **31a** or **31b**, respectively, of the chute **31**, and advances by force of gravity to the bottom end **35** (e.g., the end closer to the ground surface of the field than to the bale-forming chamber **11**) of the chute **31**. A wall, ridge or lip **37** is formed at the bottom end **35** of the second half portion **31b** of the chute **31**. There is no similar wall, ridge or lip to the ridge or lip **37** formed at the bottom end **35** of the first half portion **31a** of the chute **31**.

(18) In operation, a first bale **33a** continues to slide directly off the chute **31**, through the first half portion **31a** of the chute **31**, and onto the ground surface of the field because the chute **31** by design is substantially angled downward (see FIG. 1). Similarly, a second bale **33b** slides off the chute **31**, through the second half portion **31b** of the chute **31**, down to the bottom end **35** of the second half portion **31b** of the chute **31**, but is stopped there by the wall, ridge or lip **37** formed at the bottom end **35** of the second half portion **31b** of the chute **31**. The wall, ridge or lip **37** formed at the bottom end **35** of the second half portion **31b** of the chute **31** is configured to stop, and delay, the downward movement of the second bale **33b** until the first bale **33a** has been ejected from the chute **31** onto the ground surface of the field.

(19) Turning back to FIG. 2, an ejection actuator **39** is built into, for example, the second half portion **31b** of the chute **31**, e.g., the portion of the chute **31** with the wall, ridge or lip **37** formed at the bottom end **35** of the chute **31**. The ejection actuator **39** can be, for example, an electric linear actuator, a pneumatic cylinder, or a hydraulic cylinder, e.g., a double acting hydraulic cylinder having a moveable piston and controlled by a hydraulic control circuit (illustrated in FIG. 5, for example). However, embodiments are not limited to this configuration, and in other embodiments, the ejection actuator **39** may include a pulley system, an electric motor, a solenoid, etc.

(20) Turning back to the embodiment illustrated in FIG. 2, the ejection actuator **39** is configured for pushing the second bale **33b** stopped by the wall, ridge or lip **37** at the bottom end **35** of the second half portion **31b** of the chute **31** in the lateral direction U1. Stated differently, ejection actuator **39** is configured for pushing the second bale **33b** towards the first half portion **31a** of the chute **31** for ejection therefrom. The ejection actuator **39** may be configured to be mounted to one of the walls, e.g., a side wall **45** of the chute **31**, for example. Optionally, a cylinder support **49** may be provided between the wall **45** and the ejection actuator **39** for mounting and securing the ejection actuator **39** to the wall **45** of the chute **31**.

(21) The ejection actuator **39** can include a rod **42** and a pusher plate **47**. Movement of the rod **42** within the cylinder of the ejection actuator **39** translates to the pusher plate **47**. For example, when the rod **42** is extended (see FIG. 3), the pusher plate **47** may be driven in the direction U1 towards the first half portion **31a** of the chute **31**. Conversely, when the rod **42** is retracted, the pusher plate **47** may be driven in the direction U2 towards the second half portion **31b** of the chute **31**. The pusher plate **47** can, for example, be mounted slightly retracted from the inner surface of the wall **45** to avoid interference with bale **33b** prior to bale **33b** being pushed onto first half portion **31a**.

(22) Turning back to the operation of the bale ejection system **200**, after the first bale **33a** has

finished sliding off the chute **31** and onto the ground surface of the field, the bale ejection system **200** triggers the ejection actuator **39** to push the second bale **33b** in the lateral direction **U1** towards the first half portion **31a** of the chute **31**. FIG. **3** illustrates the ejection actuator **39** in an actuated state. In the illustrated embodiment, the piston of the ejection actuator **39** is extended in the lateral direction **U1**, thereby pushing the second bale **33b** placed on the pusher plate **47** in the lateral direction **U1** onto the first half portion **31a** of the chute **31**.

(23) After the second bale **33b** has been pushed over or slid over to the first half portion **31a** of the chute **31**, the second bale **33b** freely slides off the chute **31** and onto the ground surface of the field, similar to the sliding movement of the first bale **33a**. And, the ejection actuator **39** retracts, thereby allowing the next second bale **33a** to enter the second half portion **31b** of the chute **31**.

(24) Because the length of the chute **31** is (or may be) larger than the length of the bales **33**, the bales **33** can slide down the chute **31** and the second bale **33b** can have a certain dwell time before the next bales **33** are released from the bale-forming chamber **11** into the chute **31**. This dwell time can provide enough time for the second bale **33b** to slide over to the first half portion **31a** of the chute **31** before the next bales **33** come down from the bale-forming chamber **11** into the chute **31**.

(25) In certain embodiments, the ejection actuator **39** can be a hydraulic cylinder or an electronic actuator, for example.

(26) The ejection actuator **39** can be connected to a controller, an actuation circuit, or to other systems of the baler **10** by a hydraulic hose or a conduit, and/or by way of electric wires, cables, a bundle of wires or cables, or a wiring harness that includes various stranded or solid wires that interconnect the ejection actuator **39** with various mechanical, hydraulic, or electrical components of the baler **10**, for example.

(27) In the embodiments in which the ejection actuator **39** is a hydraulic cylinder, the bale ejection system **200** can include a hydraulic control circuit **500**, an exemplary embodiment of which is depicted in FIG. **5**, for example. The hydraulic control circuit **500** can include a hydraulic pump **502**, such as a unidirectional hydraulic pump, for example, for pumping hydraulic fluid from a fluid reservoir **504** into the hydraulic control circuit **500**. An outlet of the pump **502** and an inlet of the reservoir **504** may be connected to a directional-control valve **506**, e.g., a 4-port 2-position directional-control valve or a three-way valve, for example, operable by means of a lever or a button, or by means of an electronic control unit, having work ports connected to the ejection actuator **39**, for example, to control the hydraulic fluid flow to the hydraulic cylinder of the ejection actuator **39**. The directional-control valve **506** can be configured for switching the flow direction between the retracted position (FIG. **2**) and the extended position (FIG. **3**) of the ejection actuator **39**. This way, the ejection actuator **39** will move in the lateral direction **U1** in one setting of the directional-control valve **506** while moving in the reverse direction **U2** in the other setting of the directional-control valve **506**. Switching between both settings of the directional-control valve **506** will therefore result in reciprocation of the ejection actuator **39**, and thereby in pushing a bale **33b** in the lateral direction **U1** towards the first half portion **31a** of the chute **31**. The directional-control valve **506** may further include a third, neutral setting (not illustrated in FIG. **5**), in which hydraulic fluid flow is inhibited, for disabling the ejection actuator **39**. Alternatively, the ejection actuator **39** may be disabled by powering down the hydraulic pump **502**. The hydraulic control circuit **500** may further include means for automatic switching between the first and second setting of the directional-control valve **506**, e.g., at a fixed frequency, for example driven by an electric resonator circuit (not shown), in order to automate the reciprocating motion of the ejection actuator **39** to push a bale **33b** in the lateral direction **U1**.

(28) To facilitate management of the baling process, the bale ejection system **200** can include a control system **48**, illustrated in FIG. **6**, for example. In the illustrated embodiment, the control system includes a controller **44**, a user interface **46**, at least one bale position sensor **41**, and a hydraulic control circuit similar to the hydraulic control circuit **500** shown in FIG. **5**, for example. The controller **44** can be configured to receive signals from the operator through the user interface

46 and from sensors (e.g., the bale position sensors **41** and **43** discussed below) associated with the bale ejection system **200**. In addition, the controller **44** can be configured to send signals to the hydraulic control circuit **500** to operate valve(s) that control fluid flow to the hydraulic cylinder of the ejection actuator **39**. For example, the controller **44** can be configured to receive the signal from the bale position sensor **41** and to send a control signal to the hydraulic control circuit **500** to activate a piston of the ejection actuator **39** to extend and retract (e.g., via pumps, valves, etc.). Specifically, in the embodiment illustrated in FIG. 6, the controller **44** may send a signal to a three-way valve **64**. The signal from the controller **44** can instruct the three-way valve **64** to move into a position that enables hydraulic fluid to flow to a cap end **70** of the actuator **39**. At the cap end **70**, the pressure of the hydraulic fluid causes the piston rod **42** to extend out of the cylinder **72**, into an extended position. As a result, the pusher plate **47** may be driven in the lateral direction **U1** towards the first half portion **31a** of the chute **31**, as illustrated in FIG. 3, for example. Conversely, the controller **44** may send another signal to the three-way valve **64** that instructs the three-way valve **64** to move into a position that enables hydraulic fluid to drive the piston rod **42** back into the cylinder **72**, into a retracted position. As a result, the pusher plate **47** may be driven in the direction **U2** towards the second half portion **31b** of the chute **31**, as illustrated in FIG. 2, for example.

(29) In the embodiment illustrated in FIG. 6, the control system **48** includes a memory **58** and a processor **60**. The memory **58** may be any type of non-transitory machine readable medium for storing data and executable instructions, such as random-access memory, read-only memory, rewritable flash memory, hard drives, optical discs, and the like. The processor **60** may execute instructions stored on the memory **58**. For example, the memory **58** may contain machine readable code, such as instructions, that may be executed by the processor **60**. In some embodiments, the memory **58** and processor **60** may enable automatic (e.g., processor/memory controlled) ejection of the bales **33a**, **33b**.

(30) Embodiments of the ejection actuator **39** are not limited to a hydraulic cylinder, and in other embodiments, other types of actuator assemblies may be used. For example, an electric actuator, a mechanical actuator, or the like, may be used instead of the three-way valve **64** and hydraulic fluid to control movement of the pusher plate **47**. In certain embodiments, for example, the ejection actuator **39** may include an electric motor to move the pusher plate **47** between the first half portion **31a** of the chute **31** (FIG. 3) and the second half portion **31b** of the chute **31** (FIG. 2).

(31) In other embodiments, the bale ejection system **200** can use a mechanical, e.g., lever system, to push or move the second bales **33b** over to the first half portion **31a** of the chute **31** where there is no wall, ridge or lip at the bottom end **35** of the chute **31** that would otherwise prevent the second bales **33b** from sliding off the chute **31**.

(32) Turning back to FIG. 2, the ejection actuator **39** can be triggered in multiple ways. For example, a first sensor **41** can be arranged near the pusher plate **47** of the ejection actuator **39** (e.g., on the side wall **45** of the chute **31** or on the wall, lip or ridge **37**) and configured to sense when a second bale **33b** is loaded on the pusher plate **47** and ready to be pushed over to the first half portion **31a** of the chute **31**. A second sensor **43** can be arranged in near proximity to the bottom end **35** of the first half portion **31a** of the chute **31** and configured to sense when a bale **33a** has just been ejected from the chute **31** in order to actuate the ejection actuator **39** to push the second bale **33b** over to the first half portion **31a** of the chute **31**.

(33) In certain embodiments, the controller **44** may determine the position of the bales **33b** and **33a** based on signals received from the first sensor **41** and the second sensor **43**, respectively. The controller **44** may then send a signal to user interface **46** indicative of the bale position (e.g., a visual indication, an auditory alarm, etc.). For example, the controller **44** may send a signal to the user interface **46** indicating that the bale **33a** has just been ejected from the chute **31** based on the signal the controller **44** receives from the second sensor **43**. The signal could be a visual indication (e.g., a flashing message on display **52**, an illuminated section on the display **52**, etc.), an auditory alarm, or a combination thereof. In addition, the controller **44** may send an alert signal to alarm **54**

when the bale **33a** has not been ejected from the chute **31**. The alert signal may continue until the bale **33a** has been ejected from the chute **31** and/or the user interface **46** may show the bale **33a** position on the display **52**.

(34) Another way of triggering the ejection actuator **39** can be by monitoring a dwell time, without using the second sensor **43**. For example, the controller **44** of the ejection system **200** can sense, e.g., with the first sensor **41**, that a second bale **33b** has been placed, e.g., on the pusher plate **47** of the ejection actuator **39**, in a position to be pushed over to the first half portion **31a** of the chute **31** by the ejection actuator **39**. After the placement of the second bale **33b** is detected, the controller **44** can wait for a predetermined period of time (e.g., 5 seconds) before actuating the ejection actuator **39** to push the second bale **33b** in the lateral direction **U1** and to move the second bale **33b** over to the first half portion **31a** of the chute **31**, where there is no wall, ridge or lip (similar to the wall, ridge or lip **37**) at the bottom end **35** of the chute **31** that would otherwise keep the second bale **33b** from sliding off the chute **31**. The predetermined period of dwell time can be pre-defined or pre-programmed in the memory **58** of the controller **44** to be long enough to allow the first bale **33a** to clear the chute **31** (e.g., to be fully ejected from the chute **31** onto the ground surface of the field) before actuating the ejection actuator **39** to push the second bale **33b** over to the ejection position.

(35) The bale-shifting operating sequence of the bale ejection system **200** described herein results in the placement of the two bales (e.g., **33a** and **33b**), which are simultaneously ejected from the bale-forming chamber **11**, consecutively along the same path or the same line on the ground surface of the field. Due to the delayed ejection of the second bale **33b** until after the first bale **33a** has been ejected from the chute **31** onto the ground surface of the field, the first bale **33a** and the second bale **33b** can be ejected in a spaced arrangement relative to each other as the baler **10** moves along the field. The spaced arrangement can be longitudinal and along the same path as the moving direction of the baler **10**, due to time delay created by slowing the ejection of the second bale **33b** relative to the ejection of the first bale **33a**. The ejection of the bales **33a** and **33b** in a spaced arrangement relative to each other and along the same path on the ground surface of the field facilitates efficient collection of the bales **33** by enabling the operator of a bale bundling equipment to drive in a straight line through the field and pick up both bales, without having to deviate from the driving direction. The arrangement of the ejected bales **33** along the same path on the field allows the same bale bundling equipment to pick up both bales, as opposed to making two trips across the same path if the bales **33** were ejected side by side, in two separate lines.

(36) The improved ejection of the bales **33a** and **33b** along the same path as the moving direction of the baler **10** is illustrated in FIG. 7, for example. In FIG. 7, the movement of the baler **10** along the field is represented by an arrow A. The first bales **33a** and the second bales **33b** form a single line of bales behind one of the half portions (e.g., **31a**) of the chute **31**. As the baler **10** moves along the field carrying bales of picked up crop material, the ejection system **200** described herein ejects each of the bales **33a** and **33b** in a time-staggered manner to form a single bale collection path, which improves the efficiency of collection of the bales **33** with a single trip along a single collection path.

(37) The ejection system **200** illustrated in the figures and described above can be implemented in any hay and forage agricultural vehicle that harvests a grass type crop, including but not limited to pull-type forage harvester grass pickups, self-propelled forage harvester grass pickups, small square baler pickups, or large square baler pickups, for example.

(38) While this invention has been described with respect to at least one embodiment, the present invention can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains and which fall within the limits of the appended claims.

Claims

1. An agricultural baler, comprising: a bale chamber having a discharge outlet; and a bale ejection system configured to discharge bales from the bale chamber on a ground surface of a field, the bale ejection system including: a chute having a first half portion and a second half portion, the chute being configured to receive a first bale and a second bale in the first half portion and the second half portion, respectively; and an ejection actuator configured to push the second bale from the second half portion of the chute to the first half portion of the chute such that the first bale and the second bale are discharged in a single line on the ground surface of the field, wherein the at least one bale position sensor comprises a first bale position sensor configured to sense the presence or absence of the first bale in the first half portion of the chute and a second bale position sensor configured to sense a presence or absence of the second bale in the second half portion of the chute, wherein the ejection actuator is configured to push the second bale from the second half portion of the chute to the first half portion of the chute when the second bale position sensor senses the presence of the second bale in the second half portion of the chute and when the first bale position sensor senses the absence of the first bale in the first half portion of the chute.
2. The agricultural baler of claim 1, wherein the chute is angled downwardly from the discharge outlet toward the ground surface of the field to cause a downward sliding movement of the first bale through the first half portion of the chute.
3. The agricultural baler of claim 1, wherein the chute comprises a wall formed at an end of the second half portion of the chute.
4. The agricultural baler of claim 3, wherein the end of the second half portion of the chute is closer to the ground surface of the field than to the discharge outlet.
5. The agricultural baler of claim 3, wherein the wall is configured to prevent a downward sliding movement of the second bale through the second half portion of the chute to the ground surface of the field.
6. The agricultural baler of claim 1, wherein the ejection actuator is arranged in, on or adjacent to the second half portion of the chute.
7. The agricultural baler of claim 1, wherein the ejection actuator comprises a pusher plate configured to engage with a side portion of the second bale.
8. The agricultural baler of claim 7, wherein the ejection actuator is configured to push the second bale from the second half portion of the chute to the first half portion of the chute when the second bale position sensor senses the presence of the second bale on the pusher plate.
9. The agricultural baler of claim 1, wherein the second bale position sensor is arranged near the pusher plate.
10. The agricultural baler of claim 9, wherein the ejection actuator is configured to push the second bale from the second half portion of the chute to the first half portion of the chute after a predetermined period of time expires.
11. An agricultural vehicle comprising the agricultural baler of claim 1.
12. The agricultural baler of claim 7, wherein the pusher plate moves from a first position adjacent a side wall of the bale chamber to a second position at an intersection between the first half portion and the second half portion of the chute.
13. The agricultural baler of claim 12, wherein the first bale moves in a longitudinal direction along the first half portion, and the pusher plate moves between the first and second positions in a transverse direction that is orthogonal to the longitudinal direction.
14. The agricultural baler of claim 13 wherein the second sensor is pointed in the longitudinal direction.
15. The agricultural baler of claim 1, wherein the first half portion and the second half portion of the chute are angled downwardly at a same angle.

16. The agricultural baler of claim 1, wherein the first half portion is angled downwardly with respect to the second half portion of the chute.
 17. The agricultural baler of claim 1 further comprising a control system including a controller that receives inputs from the first and second sensors and controls the ejection actuator based upon those inputs.
 18. The agricultural baler of claim 1, wherein the first and second sensors are pointed in different directions.
 19. The agricultural baler of claim 1, wherein the first half portion and the second half portion terminate at a common edge.
-